

EC1410-Fall 2026

Problem Set 1.AI solutions

(Updated 29 September 2026)

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1. (70) The plot should look roughly like the one below.

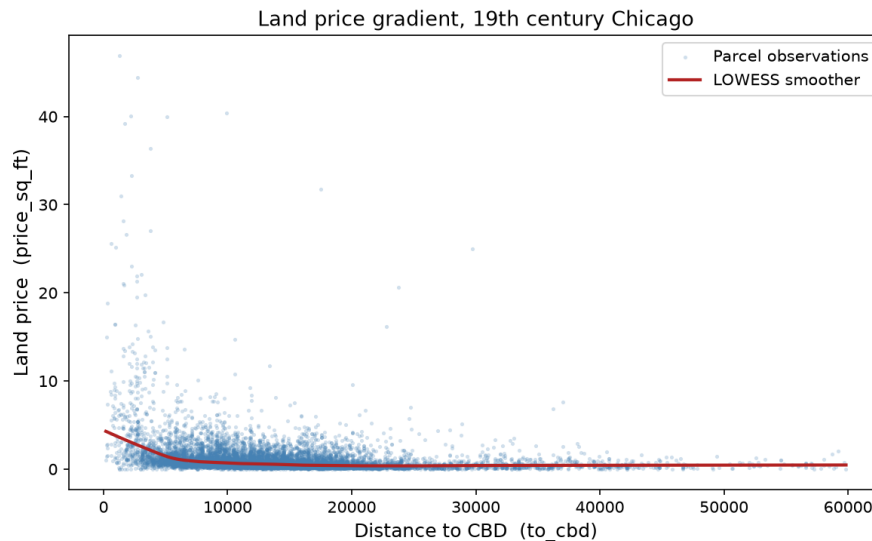


Figure 1: Land price gradient, 19th century Chicago. Each point is a parcel transaction; the red curve is a LOWESS smoother ($f = 0.20$). Source: Coury, Kitagawa, Shertzer, and Turner replication data, doi:10.7910/DVN/9TWG7D.

2. (10) **Y-axis:** price_sq_ft — transaction sale price in dollars per square foot of lot area.
3. (10) **X-axis:** to_cbd — straight-line distance from the parcel to the Chicago central business district in meters.
4. (10) The model presented in class predicts a downward sloping rent gradient just like this one. However, the model presented in class presents a linear decline, not what we see here.